

For the questions 1,2,3 consider the circuit shown in figure 1 and answer:

- 1) The equivalent resistance R_{eq} between the terminals a-b of the voltage source is: 2 Ω
- 2) The current I_o in the 6Ω resistor is: 2.22 A
- 3) The voltage (V) across 2Ω resistor is: 4.0 V

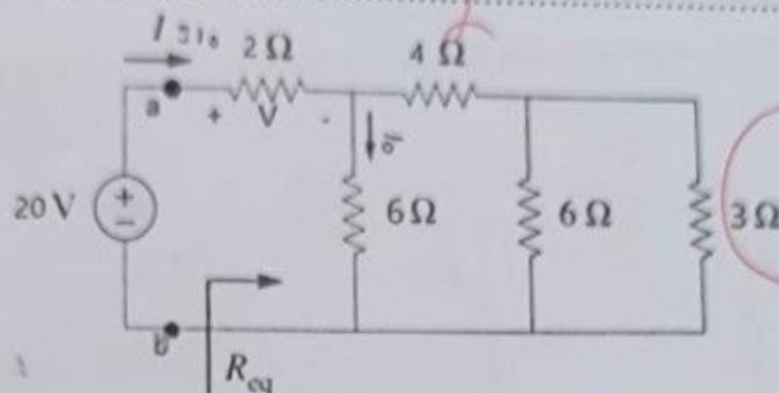


Fig.1

For the questions 4,5,6,7,8 consider the circuit shown in figure 2 and answer (use nodal analysis method):

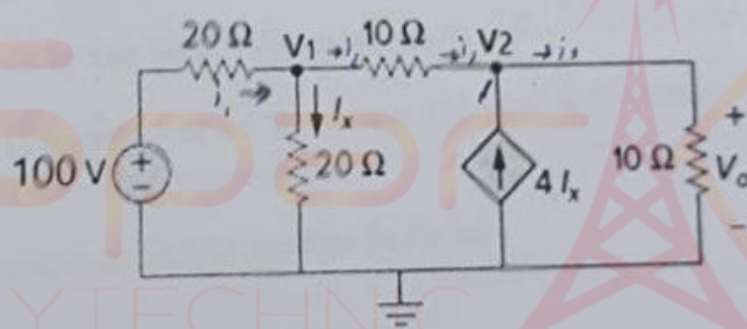


Fig.2

- 4) The value of node voltage V_1 is: 45.45 V
- 5) The value of node voltage V_2 is: 27.27 V
- 6) The value of the current (I_x) in 20Ω resistor is: 2.2725 A
- 7) The power absorbed by 10Ω resistor is: 24.36 W
- 8) Write down the nodal equations (don't simplify):

$$\frac{100}{20} - \frac{V_1}{10} - \frac{V_2}{10} + \frac{V_1}{20}$$

$$\frac{V_1 - V_2}{10} + 4I_x = \frac{V_2}{10}$$

* For the questions 9,10,11, 12,13 consider the circuit shown in figure 3 and answer:

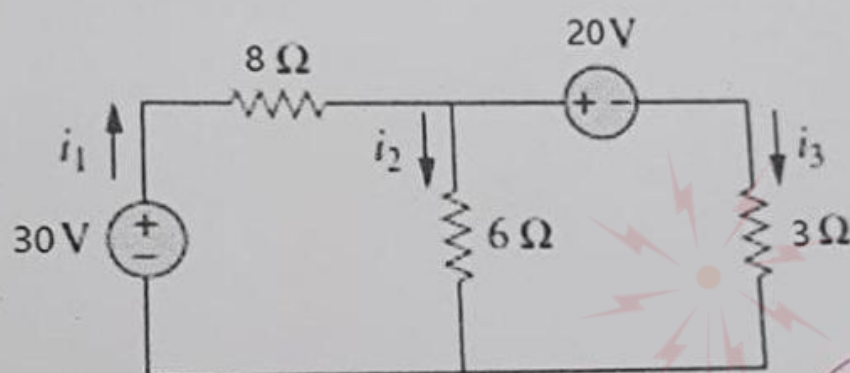


Fig.3

- 9) The current i_1 is: $\frac{30}{8} = 3.75 \text{ A}$
- 10) The current i_3 is: 6.66 A
- 11) The delivered power by the 30 V voltage source is: 112.5 W
- 12) How many joules of energy will a 6Ω resistor dissipate in 20 minutes?
 6.3360 J
- 13) The total charge entering an 8Ω resistor in 1h is:

$$1. R_{eq} = 5 \Omega \quad (4) \quad (8)$$

$$2. I_0 = 2 A \quad (2)$$

$$3. V = 8 V \quad (2)$$

$$4. V_1 = 100 V \quad (2)$$

$$5. V_2 = 150 V \quad (2)$$

$$6. I_x = 5 A \quad (2)$$

$$7. (5^2) \times 10 = 250 W \quad (2)$$

$$8. 2250 W -$$

$$\frac{V_1 - 100}{20} + \frac{V_1 - V_2}{10} + \frac{V_1}{20} = 0 \quad (2)$$

$$\frac{V_2 - V_1}{10} + 4 I_x + \frac{V_2}{10} = 0 \quad (2)$$

$$9. i_1 = 1.666 A \quad (2)$$

$$10. i_3 = -1.115 A \quad (2)$$

$$11. P_{01} = 49.8 W \quad (2)$$

$$12. W = 55444.5 \text{ Joule} \quad (2)$$

POLYTECHNIC